A Controlled Study of CNN and CBAM Attention on Synthetic SOMLAP Feature Images

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***Abstract*—Portable Executable (PE) header features permit static malware screening without executing an unknown file. We evaluate whether the Convolutional Block Attention Module (CBAM) improves a custom CNN when SOMLAP's 108 numeric attributes are arranged on a fixed 12×9 grid. The supplied CSV contains 51,408 records and 14,367 exact duplicates; identical feature vectors therefore remain within one partition or fold. Four CNN baselines, a first attention model and ten ablation configurations are evaluated. The first CNN+CBAM reaches 0.9473 Macro-F1, compared with 0.9482 for the plain CNN. After an attention-by-dropout control, the matched held-out test favors the plain CNN (0.9775 versus 0.9558 Macro-F1; McNemar $p = 1.11 \times 10^{-14}$). Five grouped folds yield 0.9823 ± 0.0040 for the plain model and 0.9809 ± 0.0042 for CBAM (Wilcoxon $p = 0.4375$). Grad-CAM and LIME describe sensitivity over synthetic feature positions; they are not interpreted as executable regions. Under this protocol, CBAM changes the error trade-off but does not improve balanced classification.**

Index Terms: malware detection, SOMLAP, convolutional neural network, CBAM, ablation, McNemar test, Grad-CAM, LIME

I. Introduction

Static screening based on Portable Executable (PE) headers avoids executing an unknown Windows file, but benign and malicious programs may still share many structural values. SOMLAP represents each sample with 108 numerical PE-header attributes. These records are tabular, whereas convolutional networks and attention modules are designed to exploit spatial neighborhoods. Mapping the attributes to a 12×9 grid retains every field and permits CNN training, but adjacency is imposed by CSV column order rather than executable structure. A high classification score therefore does not by itself establish that attention has learned a meaningful spatial dependency.

Recent malware-image studies have applied channel, spatial, self-attention and multi-head attention to byte images and malware-family benchmarks [1], [3]–[6]. Studies using SOMLAP have also reported high accuracy through ACO-based feature selection [2] and a hierarchical DNN [7]. However, their published protocols do not provide the split assignments or per-sample predictions required for a paired comparison, and they do not isolate attention from regularization in a matched CNN. The supplied SOMLAP file also contains 14,367 duplicate rows; a record-wise split could place identical feature vectors on both sides of the evaluation. It therefore remains unclear whether CBAM improves a CNN when the feature representation, backbone, dropout setting, preprocessing and data partition are controlled.

To examine that question, this study retains all 108 attributes, fits preprocessing on training data, and keeps each exact-duplicate group within one partition. Four CNN baselines establish the comparison, followed by a matched CNN+CBAM model and a two-by-two attention-by-dropout control. The selected models are compared on a held-out set and across five grouped folds using McNemar's exact test and the Wilcoxon signed-rank test. Grad-CAM and LIME are used only to inspect sensitivity over synthetic feature-grid locations. The analysis finds that CBAM changes the precision–recall balance but does not outperform the matched plain CNN on the held-out test or in mean five-fold Macro-F1.

II. Related Work

Recent malware-image research has used attention to reweight either feature channels, spatial positions or both. Wang et al. combine multiscale channel and spatial attention in MSAAM and report 99.2% family accuracy on MalImg [1]. Ravi and Alazab add multi-head attention to a CNN [3], while Huang et al. combine depthwise convolution with self-attention [4]. Fan et al. place global channel-spatial attention inside ResNet [5], and Liu et al. augment AlexNet with multi-head self-attention and Grad-CAM analysis [6]. These studies motivate channel-spatial attention, but their inputs are byte-derived images and their targets are largely malware families.

SOMLAP has primarily been studied as tabular PE-header data. Kattamuri et al. introduce the dataset and report 99.37% accuracy after ACO selects 12 of the 108 attributes [2]. Alwateer et al. report 99.11% accuracy and 98.85% weighted F1 with a hierarchical deep network [7]. Neither paper publishes the split assignments or sample-level predictions needed for a paired comparison, and neither evaluates attention on a matched CNN backbone.

The literature therefore leaves two issues unresolved for this setting. First, it is unknown whether CBAM helps when the feature representation, backbone, dropout condition, preprocessing and data partition are controlled. Second, published image explanations do not address the interpretive limit created when tabular fields are assigned artificial spatial locations. The present design tests the first issue through matched controls and treats the second by limiting Grad-CAM and LIME claims to feature-grid sensitivity.

III. Proposed Method

Each predictor is standardized with training-set statistics. The 108-value vector is reshaped to 12×9, mapped to an intensity range, resized to 96×96, and copied into three channels. This transformation retains all fields but does not create a natural image; the neighborhood relation continues to follow the original CSV order.

The custom backbone contains four convolution blocks with 32, 64, 128 and 256 filters. Batch normalization follows each convolution, and max pooling follows the first three blocks. Global average pooling converts the last feature tensor to a vector, which passes through dense layers of 256 and 128 units before the sigmoid classifier.

CBAM follows blocks 1–3 [8]. Its channel branch combines global average and maximum pooling through a shared bottleneck with reduction ratio 16. Its spatial branch concatenates channel-wise average and maximum projections and applies a 7×7 convolution. Sigmoid weights from both branches multiply the feature tensor in sequence. The attention model has 492,807 parameters, only 2,982 more than the matched plain CNN, so capacity is nearly constant in the direct comparison. The placement of the three attention modules is shown in Fig. 1.

IV. Experimental Setup

The supplied SOMLAP file contains 51,408 records, comprising 31,600 benign and 19,808 malware samples. Exact 108-value vectors define grouping units. The fixed split assigns 36,027 samples to training, 7,651 to validation and 7,730 to testing, with zero exact-group overlap. Scaling is estimated from training data; Gaussian intensity noise is confined to training, and class weights account for the 61.5:38.5 class ratio.

Models are trained with TensorFlow MirroredStrategy on two NVIDIA T4 GPUs, a global batch size of 32, Adam optimization, ReduceLROnPlateau, early stopping and a 60-epoch ceiling. Macro-F1 determines model selection. Accuracy, class-wise precision, recall and F1, macro and weighted F1, confusion matrices, ROC-AUC, PR-AUC, training time, inference time, parameter count and serialized size are also recorded.

Ten validation ablations vary attention, dropout, batch normalization, depth, augmentation and learning-rate scheduling. The selected CBAM configuration is then compared on the untouched test set with a plain CNN that differs only by the absence of attention. Five StratifiedGroupKFold partitions retain exact duplicate groups. McNemar's exact test evaluates paired test errors, while a Wilcoxon signed-rank test evaluates the five paired fold Macro-F1 values at $\alpha = 0.05$. Table I gives the fixed-split counts.

The complete 108-value vector is the split unit, so copies of one record cannot appear in two partitions. This safeguard does not establish independence by malware family, acquisition source or time because the CSV provides no stable identifier for those factors. The protocol is consequently exact-duplicate-safe rather than source-independent.

V. Results and Discussion

Among the four baselines, the plain CNN gives the highest Macro-F1 (0.9482), as shown in Table II. The first CNN+CBAM is close at 0.9473, but the similar aggregate scores conceal a different operating point: malware recall increases from 0.9416 to 0.9896 while precision decreases from 0.9315 to 0.8906. The initial result therefore supports an error-trade-off interpretation, not an improvement claim.

Figure 2 shows that the validation effect is conditional on regularization. With dropout enabled, CBAM changes Macro-F1 by -0.0191; without dropout, the change is +0.0114. CBAM without dropout is therefore selected at 0.9714 validation Macro-F1, compared with 0.9600 for the matched plain control. Removing augmentation causes the largest failure (0.4514 Macro-F1), demonstrating that the fitted model depends strongly on the training perturbation used in this experiment.

The held-out test reverses the validation ranking (Table III). The plain CNN reaches 0.9775 Macro-F1 and the selected CBAM model 0.9558. Of the discordant predictions, 275 are correct only for the plain network and 122 only for CBAM. McNemar's exact test gives $p = 1.11 \times 10^{-14}$, indicating a directional error advantage for the plain model on this test set.

The confusion counts in Table IV clarify the cost of that difference. Relative to the plain CNN, CBAM removes 88 benign false positives but introduces 241 additional malware false negatives. This is not a minor redistribution around the decision boundary. A lower threshold might recover recall, but it would need to be chosen on validation data and would change precision; no test-set threshold tuning is used here.

Model size does not explain the reversal. CBAM adds only 2,982 parameters to the 489,825-parameter plain network, and both models use the same data, feature order, normalization, image size and augmentation. Random initialization and early stopping can still affect a single fit, which motivates the paired cross-validation analysis.

The plain CNN is higher in four Macro-F1 folds and CBAM in one; Table V reports the complete metric summary. Mean Macro-F1 is 0.9823 ± 0.0040 for the plain

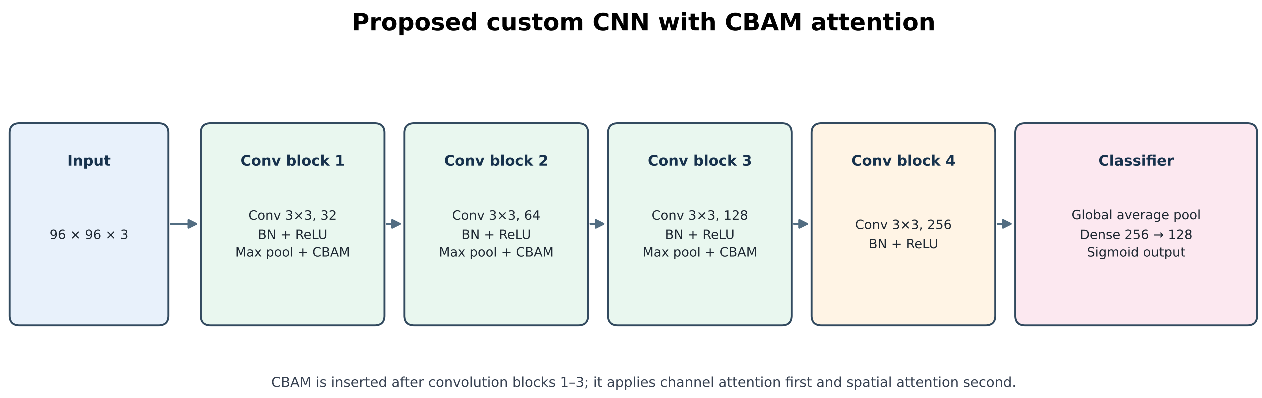

Fig. 1. Proposed custom CNN+CBAM architecture.

TABLE I
GROUPED DATA PARTITIONS USED IN THE FIXED-SPLIT EXPERIMENTS.

Partition	**Samples**	**Exact groups**	**Benign**	**Malware**
Training	36,027	25,928	22,108	13,919
Validation	7,651	5,556	4,739	2,912
Test	7,730	5,557	4,753	2,977

TABLE II
HELD-OUT BASELINE AND FIRST-ATTENTION RESULTS.

Model	**Accuracy**	**Macro-F1**	**ROC-AUC**
Plain CNN	0.9508	0.9482	0.9931
VGG16	0.9009	0.8991	0.9950
ResNet50	0.9247	0.9223	0.9875
EfficientNetB0	0.8492	0.8253	0.9670
First CNN+CBAM	0.9492	0.9473	0.9954

TABLE III
MATCHED HELD-OUT COMPARISON.

Model	**Accuracy**	**Precision**	**Recall**	**Macro-F1**
Plain, no dropout	0.9785	0.9532	0.9929	0.9775
CBAM, no dropout	0.9587	0.9794	0.9120	0.9558

TABLE IV
HELD-OUT ERROR COUNTS AT THE FIXED 0.5 THRESHOLD.

Model	**TN**	**FP**	**FN**	**TP**
Plain, no dropout	4,608	145	21	2,956
CBAM, no dropout	4,696	57	262	2,715

TABLE V
GROUPED FIVE-FOLD RESULTS REPORTED AS MEAN ± STANDARD DEVIATION.

Metric	**Plain**	**CBAM**
Accuracy	0.9831 ± 0.0038	0.9818 ± 0.0040
Malware precision	0.9719 ± 0.0085	0.9719 ± 0.0112
Malware recall	0.9848 ± 0.0025	0.9813 ± 0.0049
Malware F1	0.9783 ± 0.0048	0.9765 ± 0.0052
Macro-F1	0.9823 ± 0.0040	0.9809 ± 0.0042
Weighted F1	0.9832 ± 0.0038	0.9819 ± 0.0040
ROC-AUC	0.9982 ± 0.0005	0.9982 ± 0.0007
PR-AUC	0.9968 ± 0.0008	0.9968 ± 0.0011

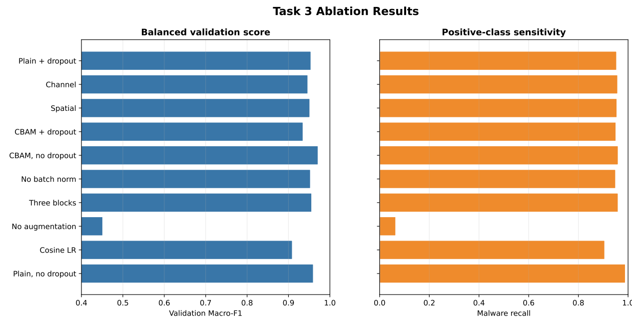

Fig. 2. Validation ablation results.

model and 0.9809±0.0042 for CBAM. The Wilcoxon result ($W = 4.0$, $p = 0.4375$) is not significant. Taken together, the fixed test and fold results support the narrower conclusion that CBAM did not outperform the matched baseline; they do not establish a stable population-level difference from five folds.

The published SOMLAP accuracies in Table VI are numerically higher, but the protocols are not matched. Kattamuri et al. use an ACO-selected 12-feature representation rather than the full 108-field grid, and Alwateer et al. use a hierarchical deep network. Neither study provides the current group assignments or predictions. Their numbers therefore locate the present result in the literature but cannot support a paired statistical claim that one method is superior.

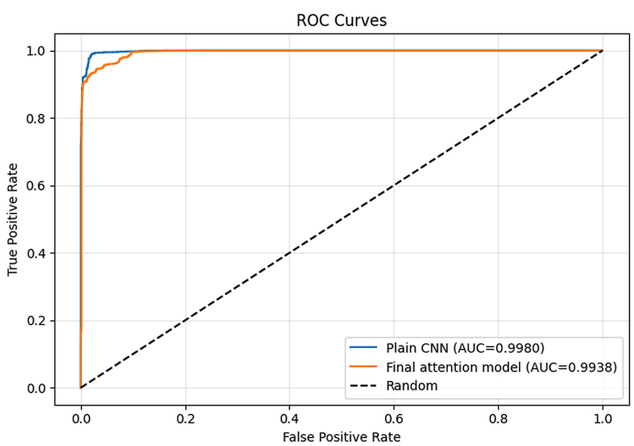

Fig. 3. Held-out ROC curves for the matched models.

Figures 3 and 4 show that both models rank samples well; the fixed-threshold confusion counts are therefore essential to the security interpretation.

Explainability and Error Analysis

The Grad-CAM examples in Fig. 5 include a correct benign sample with malware probability 0.072. The error is also benign but receives 0.992, making it a high-confidence false positive. Grad-CAM concentrates sensitivity in a small region for the correct case and spreads it more broadly for the false positive [9]. This contrast describes

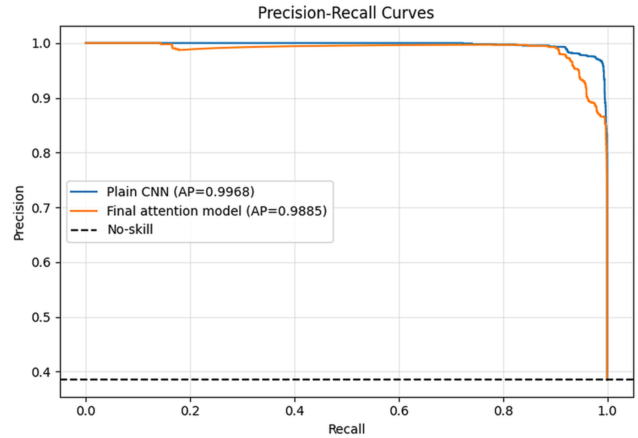

Fig. 4. Held-out precision-recall curves for the matched models.

the model's response to the two samples; it does not identify where malicious code resides.

Figure 6 presents the corresponding LIME explanations. LIME perturbs the 108 grid segments and fits a local surrogate [10]. Maxpar has the largest positive weight in both examples, while the remaining fields and weight magnitudes differ. Because resizing blends nearby feature slots and the layout follows CSV order, neither LIME regions nor Grad-CAM heatmaps are interpreted as physical executable locations or causal feature effects.

Threats to Validity

Internal validity depends on the grouping key. Exact duplicates are contained, but the available columns do not identify family, capture time or acquisition source; near-duplicates may remain across folds. Ten configurations are also examined on one validation partition. The untouched test set and grouped folds reduce selection bias but cannot replace an independently collected cohort.

Construct validity is limited by converting tabular fields to an image. The 12×9 raster preserves every value, yet adjacency comes from CSV order rather than executable structure. A convolution may exploit correlations among neighboring columns without those columns forming a meaningful spatial region. For the same reason, Grad-CAM and LIME are evidence of local model sensitivity, not localization of malicious bytes.

External validity is restricted to the supplied SOMLAP file and its extraction process. The experiments do not measure performance across malware generations, packers, collection sources or time periods.

VI. Conclusion

Under an exact-duplicate-group protocol, the compact plain CNN is the strongest model evaluated. The first CBAM model nearly matches its baseline, and the validation-selected no-dropout CBAM improves precision, but neither comparison shows a Macro-F1 advantage for

TABLE VI
SOMLAP CONTEXT. PUBLISHED ROWS ARE NOT PAIRED WITH THE CURRENT TEST SET.

Result	**Protocol**	**Accuracy**	**Weighted F1**
Plain CNN (this study)	Grouped held-out	0.9785	0.9786
CBAM (this study)	Grouped held-out	0.9587	0.9584
Kattamuri et al. [2]	ACO, 12 features	0.9937	N/R
Alwateer et al. [7]	Hierarchical DNN	0.9911	0.9885

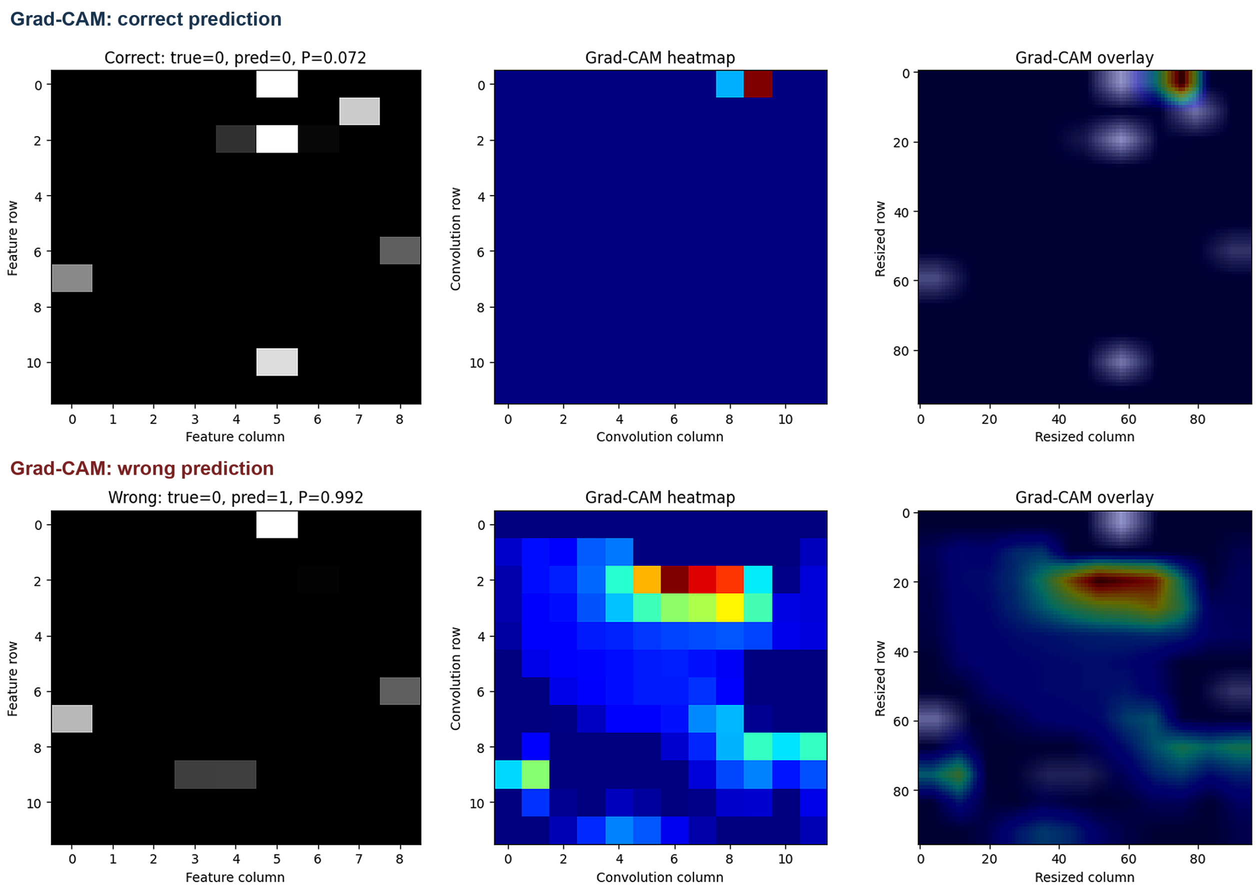

Fig. 5. Grad-CAM for the correct benign prediction (top) and the high-confidence benign false positive (bottom).

attention. On the matched held-out set, the plain CNN is higher by 0.0217 Macro-F1 and misses 241 fewer malware records. McNemar's test favors the plain model, while the five-fold Wilcoxon test does not show a significant fold-level difference.

The attention experiment remains useful because the complete two-by-two control shows that CBAM and dropout interact strongly, and the augmentation ablation identifies a major dependency of the training pipeline. The result also sets a clear interpretive boundary: a feature grid enables CNN and attention experiments, but its spatial explanations describe an engineered column layout. The central conclusion is therefore about this controlled representation and split, not about universal superiority of either CNN architecture. A subsequent experiment should select the malware decision threshold within each validation fold and lock it before testing, to determine whether CBAM's precision gain can be retained without the observed recall loss.

References

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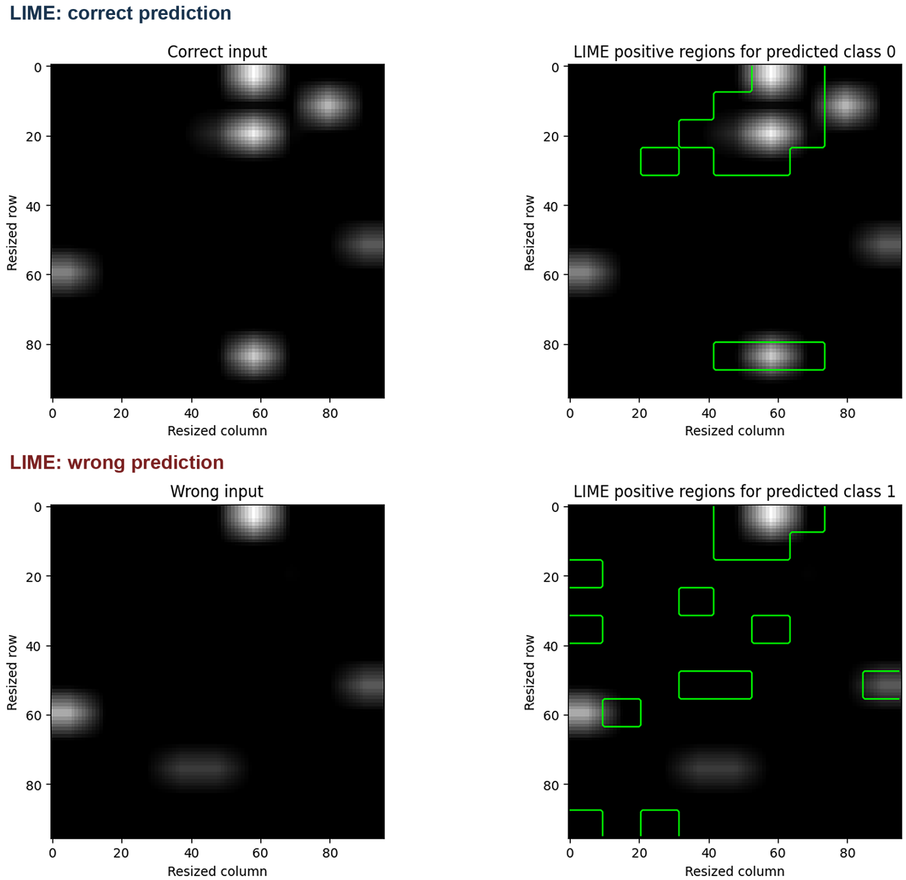

Fig. 6. LIME for the correct benign prediction (top) and the high-confidence benign false positive (bottom).

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